

Viraj Bane

Mumbai, Maharashtra, India • +91 91454 96079 • virajbane2004@gmail.com
LinkedIn: [linkedin.com/in/virubane](https://www.linkedin.com/in/virubane) • Portfolio: virrajportfolio.netlify.app • GitHub: github.com/Virajbane

SUMMARY

AI/ML graduate (2026) who has built 5+ AI apps — from a 6-agent RAG platform (93% latency cut, 33% → 82% recall@6) to a multilingual NL-to-SQL engine — now looking to bring that hands-on build experience to an entry-level engineering team.

EDUCATION

Viva Institute of Technology, B.E. in Computer Science & Engineering (AI & ML) **CGPA: 8.1, Aug 2023 – Jun 2026**
Vidyavardhini Bhausaheb Vartak Polytechnic, Diploma in Computer Engineering **78%, Dec 2020 – Jun 2023**

SKILLS

Languages: Python, SQL, JavaScript, Java
AI/ML: Machine Learning, Deep Learning, LLMs, RAG, NLP, Transformers, Transfer Learning, Fine-tuning, MLOps
Frontend: React.js, Next.js, Tailwind CSS
Backend: FastAPI, Node.js, LangChain, LangGraph, REST APIs, JWT Authentication
Libraries: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Databases: MongoDB, MySQL, Redis, Qdrant, ChromaDB, Supabase
Tools & Platforms: Git, GitHub, Docker, Hugging Face, Ollama, Vercel, Google Colab, n8n

PROJECTS

Adaptive-RAG – Enterprise Multi-Agent RAG Platform [[GitHub](#)] May 2026 – Jun 2026

- Architected a 6-agent pipeline (Rewriter, Planner, Tool, Grader, Answer, Critic) **orchestrated with LangGraph**, with deterministic validation and routing safeguards for each LLM call to catch routing and hallucination failures missed by the base model.
- Designed a hybrid retrieval engine fusing Qdrant dense vector search and BM25 keyword search using Reciprocal Rank Fusion (RRF), refined with a Cross-Encoder reranker and a Grader agent for noise filtering.
- Migrated PDF ingestion to Docling for structure-aware parsing and table extraction, **improving retrieval Recall@6 from 33% to 82% and MRR from 0.111 to 0.778** on a golden-set benchmark.
- Cut repeat-query latency by 93% (28s → 2s) with a Redis caching layer**; secured the FastAPI backend with JWT authentication and rate limiting.
- Built a RAG evaluation harness using **Ragas** (30-question golden set, dual faithfulness/relevance LLM-judge scoring) that surfaced and fixed 3 production bugs, including a silently broken hybrid search and a hallucination-detection gap in the answer-validation agent.

AI DB Agent – Natural Language Interface for SQL & NoSQL Databases [[GitHub](#)] Nov 2025 – Dec 2025

- Built a full-stack NL-to-query interface spanning 5 database systems (MongoDB, PostgreSQL, MySQL, Redis, and Supabase) using FastAPI, Next.js, and Ollama (Qwen2.5-Coder 7B).
- Implemented multilingual query support for English, Hindi, and Marathi using hybrid intent classification for accurate query generation.
- Accelerated Ollama inference handling, **cutting cold-start latency from ~3.5 minutes to ~12 seconds response time (94% reduction)**, while enforcing schema-aware generation with SQL injection prevention.

EDU.AI – AI-Powered Classroom Learning Platform [[GitHub](#)] Oct 2024

- Developed an AI-powered learning platform during **Mumbai Hack 2024 — a Guinness World Record-recognized hackathon** — integrating 3 ML-driven features: lecture summarization, transcript analysis, and keyframe extraction via Python microservices.
- Designed a full-stack platform with Next.js, Tailwind CSS, Supabase, Prisma, and Google OAuth, supporting secure authentication and scalable multi-user access.
- Delivered personalized study routines and real-time learning analytics, translating raw lecture content into structured exam preparation material for students.

EXPERIENCE

Python Developer Intern – Branding Catalyst Jul 2022 – Aug 2022

Vasai, Maharashtra

- Wrote Python automation scripts to streamline routine tasks and improve workflow efficiency.
- Debugged, tested, and optimized Python applications to improve reliability and performance.

CERTIFICATIONS & ACHIEVEMENTS

- Published 2 fine-tuned NLP models on Hugging Face Hub and built 2 AI automation agents using n8n.
- Published a Research Paper: “EYES.AI: Vision Assistance Using AI” – NCRENB 2026, VIVA-TECH IJRI (ISSN: 2581-7280).
- NPTEL Deep Learning – IIT Ropar, Elite Badge, 12-week certification (Jul – Oct 2025).
- 100-Hour Cyber Security Training (Grade B) – Atos Prayas Foundation, ICT Academy, Viva Institute of Technology (Jul – Aug 2025).